Department of Physics, Princeton University

Graduate Preliminary Examination
Part I

Thursday, May 9, 2024
9:00 am - 12:00 noon

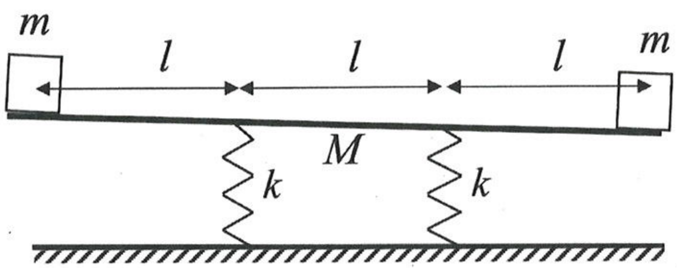
Answer TWO out of the THREE questions in Section A (Mechanics) and TWO out of the THREE questions in Section B (Electricity and Magnetism).

Work each problem in a separate examination booklet.

Be sure to label each booklet with your name, the section name, and the problem number.

Section A. Mechanics

1. **See-Saw**



A playground see-saw is supported by two vertical springs with equal spring constants k. Assume that the children can be represented by equal point masses m at each end of a long uniform see-saw rod that has mass M and length 3ℓ. The distances between the springs and from each child to the closest spring are all equal to ℓ. Consider small oscillations of this system in the plane of the drawing. You may assume that the springs each only move up and down (i.e. no bending). Find the frequencies of the two normal modes of this system.

2. **A Rolling Ball**

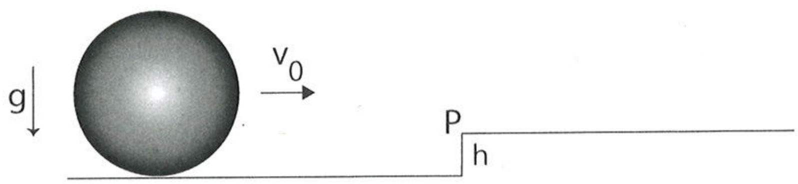


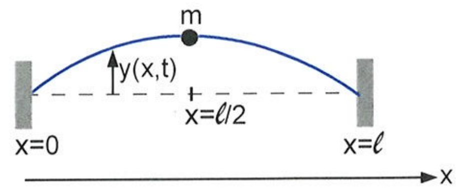
Figure 1: Bowling ball encounters a step. The gravitational acceleration $\mathbf{g}$ points down as shown.

A solid bowling ball of radius a and mass M rolls without slipping on a horizontal plane with a center-of-mass (CM) velocity $\mathbf{v}_0$. Let its rotational inertia about its CM be I_0. The ball hits a step of height h (see Fig. 1). The collision is inelastic and very short in duration.

a) Find the angular velocity of the ball ω_p immediately *after* impact with the corner P.

b) Assuming its initial kinetic energy is high enough, the ball pivots over the step without slipping and resumes rolling. Find the final velocity.

3. **Point mass on a string**



Consider a string of length ℓ with uniform mass density σ and tension τ. Its endpoints at $x = 0$ and $x = \ell$ are held fixed. A point mass m is attached at the center of the string, at $x = \ell/2$. The mass can only move in the vertical direction (it cannot slide along the string). Gravity can be neglected. The purpose of this problem is to find the normal mode frequencies for the small vertical oscillations $y(x,t)$ of the string in the presence of the mass. You may assume that $y(x,t)$ obeys the standard wave equation.

(a) The wave equation should be supplemented with the boundary conditions at $x = 0$, $x = \ell$, and with appropriate "matching conditions" on $y(x,t)$ and its derivative at $x = \ell/2$. Determine the form of these matching conditions.

(b) We may search for solutions of the wave equation in the normal mode form

$$y(x,t) = \rho(x)\sin(\omega t + \phi)$$

Use the boundary conditions and matching conditions derived above to find the possible oscillation frequencies ω. It is convenient to consider separately the modes for which the string has a node at $x = \ell/2$ ("odd modes") and those for which it does not have a node at $x = \ell/2$ ("even modes"). For the odd modes, determine the form of the frequencies explicitly. For the even modes, find the transcendental equation that determines the allowed frequencies.

(c) Find the form of the even mode frequencies in the limit $m/(\sigma\ell) \to 0$.

(d) In the opposite limit $m/(\sigma\ell) \to \infty$, find an approximate expression for the lowest oscillation frequency, to leading non-trival order in the small quantity $\sigma\ell/m$.

Section B. Electricity and Magnetism

1. **Magnetic Pressure**

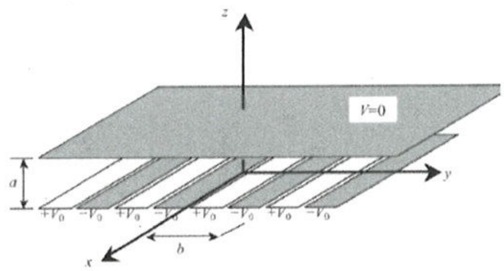
   Consider an infinitely long cylinder of incompressible conducting plasma with an axial current density $J_z = J(r)$ and a resulting azimuthal magnetic field $B_\phi = B(r)$. The coordinate r is the radial coordinate with respect to the axis of the cylinder. Assume the current density is confined within a finite radius R of the cylinder. The integral of the current density over the cross-sectional area of the cylinder is equal to the total current I.

   a) Determine $B_\phi = B(r)$ for $r \geq R$ in terms of I and r.

   b) Now assume the current density is constant within the cylinder, $J_z = J_0$. Find $B(r)$ for $r < R$.

   c) The Lorentz forces on the current will try to "pinch" the plasma and squeeze it radially inward. These pinching forces are balanced by hydrostatic pressure gradient forces, which in this case keep the plasma at a constant density. Find an expression for the hydrostatic pressure, $p = p(r)$, inside the plasma that will generate the forces to keep a uniform current density of part b), and sketch the dependence of p on r. You can assume that the pressure is zero at the surface of the plasma.

2. **An electrostatics problem**



A device is made of two very large conducting parallel planes separated by a distance a in the z direction. The distance a is much smaller than the size of the planes, which you may take to be infinite in this problem. The plane at $z = a$ is grounded, and you can set its electrostatic potential to be $V = 0$. The plane at $z = 0$ is made of parallel strips of width $b/2$, see the figure. The voltage on the strips is alternating between $+V_0$ and $-V_0$. You may ignore the gaps between the strips.

Find the electrostatic potential $V(y, z)$ everywhere between the plates. Your final result can be left in the form of an infinite series. [Hint: Use separation of variables, and also note that the voltage of the lower plane can be regarded as a periodic function of y with period b.]

3. **Plasma Waves**

A "tenuous plasma" consists of free electrons with mass m and charge e. There are n electrons per unit volume. Assume that the electron density is uniform and that interactions between the electrons may be neglected. Electromagnetic waves (frequency ω and wave number k) are incident on the plasma.

(a) Find the conductivity σ of the plasma as a function of ω.

(b) Find the dispersion relation; i.e. find the relation between k and ω.

(c) Find the index of refraction as a function of ω. What does it tell you about the speed of wave propagation in the plasma? The plasma frequency is defined as $\omega_p^2 = ne^2/m\epsilon_0$ (in SI units). What happens if $\omega < \omega_p$?

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Graduate Preliminary Examination
Part II

Friday, May 10, 2024
9:00 am - Noon

Answer TWO out of the THREE questions in Section A (Quantum Mechanics) and TWO out of the THREE questions in Section B (Thermodynamics and Statistical Mechanics).

Work each problem in a separate booklet. Be sure to label each booklet with your name, the section name, and the problem number.

Section A. Quantum Mechanics

1. **A Spinning Particle in a Magnetic Field**

   A spin 1/2 particle of intrinsic magnetic moment $\vec{\mu} = \gamma\vec{S}$ has spin up along the z direction. At time $t = 0$ a magnetic field $\vec{B} = B\hat{y}$ is turned on.

   a) At $t = T$ what is the probability that the spin is down?

   b) At $t = T/2$ another experimentalist measures the z-component of the spin but does not tell you the result. What is the probability that your subsequent measurement at $t = T$ will find the spin to be down?

   c) Consider now a series of such intermediate measurements at times $t = kT/N$, where k and N are integers, $N \gg 1$ and $k = 1...(N-1)$. You record the result of the last measurement at T. Assuming that $\gamma BT/N \ll 1$, what is the probability that your final measurement at $t = T$ will find the spin to be down?

2. **Two Hydrogen Atoms**

Consider two hydrogen atoms with their nuclei held at fixed positions such that the displacement $\vec{R}$ between them is large compared to the ground-state size of the atoms. Treat the Coulomb interaction as instantaneous (no retardation), neglect nuclear spins and nuclear motion, and neglect any contribution of the electron spins to the energy (no spin-orbit interactions). The appropriate basis set for discussing the energetics of this system is spanned by states of the type $\psi^{(1)}_{n\ell m}(\vec{r}_1)\psi^{(2)}_{n'\ell' m'}(\vec{r}_2)$ where each electron is in a single-atom eigenstate about its nucleus. Take the z axis to lie along the displacement $\vec{R}$ between the two nuclei. Under these conditions, the dominant interaction between the two atoms comes from the electric dipole-dipole interaction :

$$U_{dipole}(\vec{R}) = \frac{1}{4\pi\epsilon_0 R^3}\left[\vec{d}^{(1)} \cdot \vec{d}^{(2)} - 3d_z^{(1)} d_z^{(2)}\right]$$

where $\vec{d}^{(n)}$ is the dipole operator for the electron bound to nucleus n. The two electrons can of course be in either a state of total spin $S = 0$ or $S = 1$; in what follows, let the two electrons always have total spin $S = 0$.

(a) The ground state energy of this pair of atoms depends on R at large R as

$$E_0(r) = E_0(\infty) + A_0 R^{-\delta_0} + \ldots .$$

Explain what process determines the value of the exponent δ_0 and give the value of δ_0.

(b) Give an order of magnitude estimate for the constant A_0 and give a general argument for its sign.

(c) The energy of the lowest energy electronic *excited* eigenstates of this pair of atoms depends on R at large R as

$$E_1(R) = E_1(\infty) + A_1 R^{-\delta_1} + \ldots .$$

What is the exponent δ_1? Give an order of magnitude estimate of A_1. What is the sign of A_1? What is the full wavefunction of this excited eigenstate at leading order in large R?

3. **Degeneracies of the 2D harmonic oscillator.**

A spinless particle of mass m is placed into a 2D harmonic oscillator potential with classical frequency ω:

$$H = \frac{\hbar\omega}{2}\left(P_x^2 + P_y^2 + X^2 + Y^2\right) = \frac{\hbar\omega}{2}\left[-\frac{1}{R}\frac{\partial}{\partial R}R\frac{\partial}{\partial R} - \frac{1}{R^2}\frac{\partial^2}{\partial\phi^2} + R^2\right],$$

where, in order to avoid clutter, we define the dimensionless operators:

$$X = x\sqrt{m\omega/\hbar}\,, \qquad Y = y\sqrt{m\omega/\hbar}\,, \qquad P_x = p_x/\sqrt{\hbar m\omega}\,, \qquad P_y = p_y/\sqrt{\hbar m\omega}\,,$$

as well as the polar coordinates (R, ϕ) obeying $X = R\cos\phi$ and $Y = R\sin\phi$.

(a) What are the energy levels of H and their degeneracies?

(b) What is the unnormalized wavefunction $\psi_0(R, \phi)$ for the ground state?

(c) The potential is rotationally symmetric, so the angular momentum

$$L = \hbar(XP_y - YP_x)$$

is conserved. However, the conservation of L by itself cannot explain the presence of all degeneracies in the spectrum of H, so this suggests that there are more conserved quantities. Indeed, here are two more:

$$J_1 = (\hbar/4)(X^2 - Y^2 + P_x^2 - P_y^2)\,, \qquad J_2 = (\hbar/2)(XY + P_xP_y)\,.$$

It is straightforward to check that both J_1 and J_2 commute with H. Interestingly, J_1, J_2, and L have the same commutation relations with each other as the 3D angular momentum operators! In the usual normalization:

$$[J_1, J_2] = i\hbar J_3\,, \qquad [J_2, J_3] = i\hbar J_1\,, \qquad [J_3, J_1] = i\hbar J_2\,,$$

where $J_3 = \alpha L$ for some numerical constant α. Calculate the value of α.

(d) Consider a wavefunction of the form

$$\psi_n = NR^n e^{in\phi}\psi_0\,,$$

where n is a non-negative integer and N is a normalization constant that you do not need to determine. Check that this wavefunction is a simultaneous eigenfunction of H and L and compute the corresponding eigenvalues.

(e) Use part (d) and your knowledge of angular momentum quantization to decide whether the existence of the three conserved quantities J_1, J_2, and J_3 fully explains the degeneracies you calculated in part (a).

Section B. Statistical Mechanics and Thermodynamics

1. **Temperature of the Atmosphere**

   Here you will model Earth's atmosphere as an adiabatic ideal gas. This assumes that, due to the weather, the gas rises and descends fast enough that thermal conduction through the gas is negligible, so the atmosphere is adiabatic and is at mechanical equilibrium. Assume the pressure $P(z)$ and temperature $T(z)$ depend on the vertical position z (the height), and are uniform over all horizontal positions at a given height z. The temperature and pressure at the surface ($z = 0$) are T_0 and P_0. The heat capacity of a mole of this gas at fixed volume is C_V.

   (a) What is $T(P)$, the temperature as a function of the pressure for this adiabatic gas, given C_V, T_0 and P_0?

   (b) If this gas has mass M per mole, what is $T(z)$, the temperature as a function of the height? Assume the atmosphere is shallow enough so the acceleration of gravity g is a constant.

   (c) Assuming the atmosphere is purely molecular nitrogen, N_2, with molecular weight 28, what is dT/dz in degrees K per kilometer (give your best estimate of the magnitude, and give the *sign*) at the surface, where we live?

2. **Particles in a Box**

First consider a single free non-relativistic particle of mass m confined to a three-dimensional cube of volume $V = L^3$. The particle has no internal degrees of freedom. Let $Z_1(mT)$ denote the quantum partition function for this system at temperature T (where the partition sum is taken over the discrete energy levels of this system).

(a) Working in the classical (high temperature) regime, show that $Z_1(mT) \approx V/\lambda^3$ with a suitably defined de Broglie wavelength $\lambda(mT)$. Use this result to obtain the classical energy and heat capacity at fixed volume of this single-particle system.

(b) What (roughly) is the temperature at which this classical approximation breaks down?

(c) Now consider a system consisting of two identical, non-interacting such particles in the same box. The particles either have no internal degrees of freedom in the case of bosons, or these degrees of freedom are held fixed in the case of fermions (e.g. both spins "up"). The two-particle partition function $Z_2(mT)$ contains the effects of identical-particle statistics. Show that the exact free boson and free fermion two-particle quantum partition functions $Z_2(mT)$ can in fact be expressed in a simple way in terms of the exact one-particle quantum partition functions $Z_1(mT)$ and $Z_1(mT/2)$.

(d) Using the classical approximation $Z_1 = V/\lambda^3$ derived in the first part of this problem, calculate the leading high-temperature contribution to the energy E and the heat capacity C_V due to Bose or Fermi statistics in this two-particle system.

3. **Hydrogen Recombination**

In this problem we investigate the formation of hydrogen atoms in the early universe. Although the binding energy of hydrogen is 13.6 eV, the majority of protons and electrons did not become bound into atoms until the temperature of the neutral primordial plasma cooled to about 0.3 eV.

In the following we make four assumptions:

- The hydrogen atom has no bound states apart from its ground state.
- We ignore other bound complexes that might be formed, e..g, hydrogen ions and molecules.
- All interactions among hydrogen atoms protons and free electrons are ignored (apart from the fundamental process of atom formation).
- Everything is in thermal equilibrium.

There are two questions:

a) Assume that at $T = 0.3$ eV half of the protons have a bound electron. From this information calculate the densities (in units of particles per cubic meter) of free electrons, free protons, and hydrogen atoms. (*Depending on how you approach this problem, Stirling's formula might be useful:* $N! = \sqrt{2\pi N} N^N e^{-N}, \quad N \gg 1$)

b) At $T = 0.3$ eV what is the number of density of photons relative to that of the Hydrogen atoms? An order of magnitude estimate will suffice for this part of the problem.